

CHE 201 Spring 2020 Midterm 1

Roxanna Mendoza

TOTAL POINTS

59 / 100

QUESTION 1

Question 1 20 pts

1.1 a 8 / 10

- 0 pts Correct
- 2 pts first is incorrect
- ✓ - 2 pts second is incorrect
- 2 pts third is incorrect
- 2 pts fourth is incorrect
- 2 pts fifth is incorrect

1.2 b 5 / 5

- ✓ - 0 pts Correct
- 2 pts gauge pressure not converted to absolute
- 2 pts T not converted to K
- 1 pts P1 is wrong

1.3 C 5 / 5

- ✓ - 0 pts Correct
- 5 pts incorrect

QUESTION 2

Question 2 50 pts

2.1 a 3 / 3

- ✓ - 0 pts Correct

2.2 b 8 / 8

- ✓ - 0 pts Correct
- 4 pts Pv incorrect
- 4 pts Tv incorrect
- 7 pts States not labeled in each
- 2 pts pv state 1 and state 3 incorrect
- 1 pts tv state 3 incorrect
- 2 pts both graphs state 1 is wrong
- 2 pts both graphs state 2 is wrong

2.3 C 5 / 5

- ✓ - 0 pts Correct

2.4 d 2 / 2

- ✓ - 0 pts Correct

2.5 e 12 / 15

- ✓ - 0 pts Correct
- 3 pts v1 is incorrect or not calculated
- 3 pts v2 is incorrect
- 2 pts not multiplied with mass
- 2 pts units are not converted
- 15 pts not done
- 3 Point adjustment
- to need to multiply specific volume with mass to calculate volume. Not divide. Also mixed up v1 and v2

2.6 f 4 / 15

- 0 pts Correct
- 3 pts u1 is incorrect or not calculated
- 3 pts u3 is incorrect or not calculated
- ✓ - 15 pts incorrect
- 2 pts x is wrong
- 2 pts u not multiplied with mass
- 2 pts calculation error

+ 4 Point adjustment

- Q and U are not the same.

2.7 g 1 / 2

- 0 pts Correct
- 2 pts incorrect or not done
- 0.5 pts s1 is wrong
- ✓ - 1 pts s3 is wrong
- 0.5 pts s2 is wrong

QUESTION 3

Question 3 30 pts

3.1 a 2 / 2

✓ - 0 pts Correct

3.2 b 2 / 2

✓ - 0 pts Correct

- 0.5 pts ideal gas is missing

- 0.5 pts closed system is missing

3.3 c 0 / 3

- 0 pts Correct

- 2 pts Z ideal and Znotideal are not correct

- 1 pts Z for not ideal gas is not correct

✓ - 3 pts wrong

3.4 d 2 / 3

- 0 pts Correct

- 3 pts incorrect

✓ - 1 pts line should be not linear

- 1 pts P and V are not marked

3.5 e 0 / 10

- 0 pts Correct

- 1 pts calculation error

✓ - 10 pts incorrect or not done

💬 This is not constant P system

3.6 f 0 / 10

- 0 pts Correct

✓ - 10 pts incorrect or not done

- 1 pts calculation error

NAME: Roxanna MendozaPlease fill in answers in the space supplied and show all your work to receive full credit.
Useful Tables at end of Exam.1. a) (10 pts) State whether the following statements are **True or False**F An ideal gas at a given state expands to a fixed final volume at constant pressure or at constant temperature. Work is smaller in constant pressure case.F A two-phase liquid-vapor mixture with equal volumes of saturated vapor and saturated liquid has a quality of less than 0.5.T As temperature decreases from the critical temperature, the values of the specific volume of saturated vapor and saturated liquid moves apart from each other.F Internal energy is an intensive property.F Specific volume of compressed liquid can be approximated to specific volume of saturated liquid at the same pressure.

b) (5 pts) The gage pressure of an automobile tire is measured to be 30psig before a trip and 32psig after the trip at a location where the atmospheric pressure is 14psia. Assuming the volume of the tire remains constant and the air temperature before the trip is 25°C, determine the air temperature in the tire after the trip. (Air is ideal gas)

$$\frac{P_1}{T_1} = \frac{P_2}{T_2}$$

$$P_{\text{gage}} = P_{\text{absolute}} - P_{\text{atm}} \quad 25^\circ\text{C} \rightarrow 298\text{K}$$

$$30\text{ psig} + 14\text{ psia} = 44\text{ psig}$$

$$\frac{44}{298} = \frac{46}{T_2}$$

$$32\text{ psig} + 14\text{ psia} = 46\text{ psig}$$

$$T_2 = 311.54\text{ K}$$

c) (5pts) What would be the specific volume of saturated solid carbon dioxide at the same temperature, greater or less than saturated vapor value? Why?

$$\text{Specific volume} = \frac{\text{Volume}}{\text{mass}}$$

The specific volume of saturated solid carbon dioxide at the same temperature would be less than saturated vapor value because the volume does not change only the mass does which then would indicate a bigger denominator then implying smaller specific volume value.

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2. (50 pts) 5kg of water at contained in a piston-cylinder assembly undergoes two processes in series from an initial state where the temperature is 120°C and 10 bar.

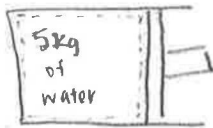
- **Process 1-2** System is heated at constant pressure to a saturated vapor state.
- **Process 2-3** System is cooled at constant volume until it reaches to 120°C.

Kinetic and potential effects can be ignored.

$W=0$ for constant volume

- (3 pts) Draw a schematic of the system and label the system boundary.
- (8 pts) Carefully sketch the process on a:
 - P-v diagram (Pressure-specific volume)
 - T-v diagram (Temperature-specific volume)
- (5 pts) State the engineering model
- (2 pts) Write the simplified energy balance
- (15 pts) Calculate the work transferred in kJ. Is work done on the system or by the system?
- (15 pts) Calculate the heat transferred in kJ. Is heat transferred to the system or from the system?
- (2 pts) Calculate the mass of vapor in each state in kg $m_g = x m_g$

Part A:

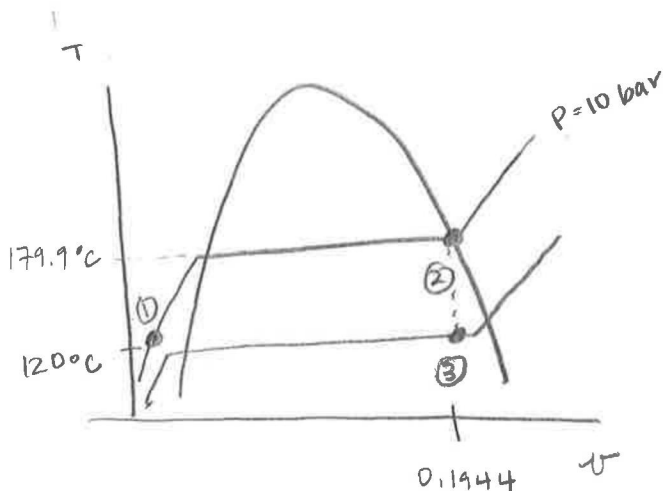


Part C:

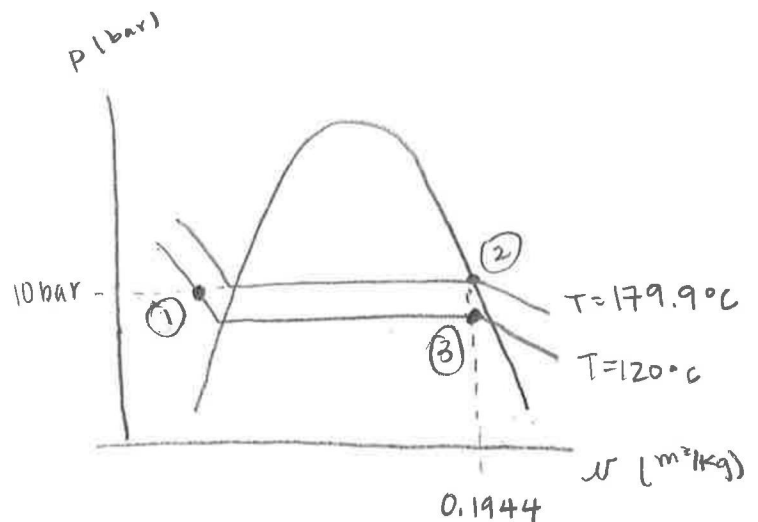
Engineering Model

- closed system
- $\Delta KE = \Delta PE = 0$
- work done is volume change

Part B:



$$v_2 = v_3$$



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$$\Delta E = Q + W$$

$$\Delta P \overset{0}{\cancel{E}} + \Delta K \overset{0}{\cancel{E}} + \Delta U = Q + W$$

$$\Delta U = Q + W$$

Part E:Process 1 → 2: $P = 10 \text{ bar}$; $v_g = 0.1944$; $T = 179.9^\circ\text{C}$;

$$u_g = 2583.4$$

$$V_1 = \frac{0.1944}{5 \text{ kg}} = 0.03888 \text{ m}^3$$

$$V_2 = \frac{1.0603 \times 10^{-3}}{5 \text{ kg}} = 2.1206 \times 10^{-4}$$

$$W = \int_{V_1}^{V_2} P dV = P(V_2 - V_1)$$

$$W = (10,000 \text{ kPa})(V_2 - 0.03888 \text{ m}^3)$$

$$W = -386.68 \text{ kJ on the system}$$

$$2583.4 - 503.50 = 2080.1$$

$$Q = 2080.1 \text{ kJ}$$

Process 2 → 3: $V = \text{constant}$, $v = 0.1944 \text{ m}^3/\text{kg}$

$$W = 0 \text{ kJ}$$

$$\Delta U = Q$$

$$v = (1-x)v_f + x v_g$$

$$v = (1-x)(1.1273 \times 10^{-3}) + x(0.1944)$$

$$v = 1.1273 \times 10^{-3} - 1.1273 \times 10^{-3}x + 0.1944x$$

$$v = 1.1273 \times 10^{-3} - 0.19327x$$

$$v =$$

① compressed liquid $x < 0$ ② saturated-vapor state $x = 1$

③ liquid-vapor mixture

Part G:

State 2 vapor mass

$$m_2 = x m_T \Rightarrow m_2 = (1)(5 \text{ kg})$$

$$m_2 = 5 \text{ kg of vapor}$$

10 bar	10^5 N/m^2	1 Pa	0.001 kPa = 10,000 kPa
1 bar	1 N/m ²	1 Pa	

State 1 vapor mass

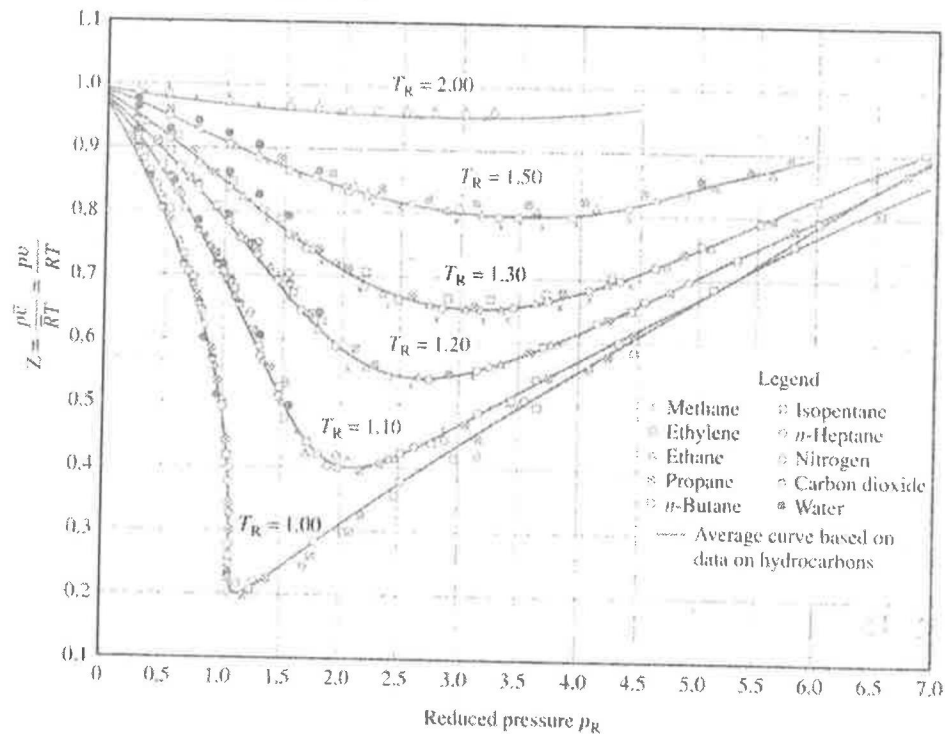
$$m_1 = 0 \text{ kg of vapor}$$

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3. (30 points) A piston cylinder device initially contains 0.5m^3 of air at 80kPa and 200°C . The air is now compressed to 0.1m^3 in such a way that the temperature inside the cylinder remains constant. The air is modeled as an ideal gas and kinetic and potential energy effects are negligible.
- (2 pts) Draw a schematic of the system and label the system boundary.
 - (2 pts) State the engineering model
 - (3 pts) Write the simplified energy balance and calculate compressibility factor (Z) for ideal gas and not ideal gas state 1.
 - (3pts) Draw P-v diagram (Pressure-specific volume)
 - (10 pts) Determine the work (kJ)
 - (10 pts) Determine the heat (kJ)

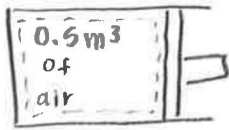


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Part A:



Part B:

Engineering Model:

- ① closed system
- ② $\Delta KE = \Delta PE = 0$
- ③ ideal gas $PV = nRT$
- ④ $T = \text{constant}$

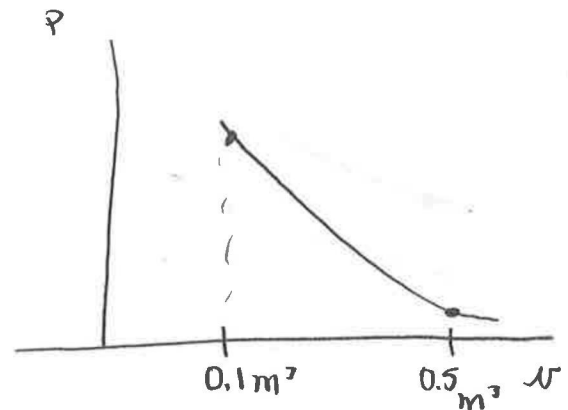
Part C:

$$\Delta E = Q + W$$

$$z = \frac{pV}{RT} \quad 200^\circ\text{C} \Rightarrow$$

$$z = \frac{(900 \text{ kPa})(0.5 \text{ m}^3)}{(8.314 \text{ kJ/kmol} \cdot \text{K})(473 \text{ K})} = 0.0102$$

Part D:



Part E:

$$\Delta h = Q + W$$

$$h = 199.97$$

$$199.97 = Q + 32$$

$$Q = 167.97 \text{ kJ}$$

$$W = \int P dv$$

$$W = 900 \text{ kPa} (0.5 - 0.1)$$

$$W = 32 \text{ kJ} \quad \text{By the system}$$

